However the secondary images, inside the Einstein ring, move in an unexpected manner: They appear to emerge from the right edge of the shadow, move outward into the annulus between the shadow and the Einstein ring, swing leftward around the shadow, and descend back toward the shadow's edge.

You can understand this by going back to the upper drawing in Figure 8.3. The right ray passes near the black hole, so the right stellar image is near its shadow. Earlier in time, when the camera was further leftward, the right ray had to pass even closer to the black hole in order to bend more strongly and reach the camera, so the right image was very close to the edge of the shadow. By contrast, earlier in time, the left ray passed rather far from the hole and so was nearly straight and produced an image rather far from the hole.

Now, if you're ready, think through the subsequent motions of the images, depicted in Figure 8.4.

Lensing by a Rapidly Spinning Black Hole: Gargantua

The whirl of space generated by Gargantua's very fast spin changes the gravitational lensing. The star patterns in Figure 8.1 (Gargantua) look somewhat different from those in Figure 8.4 (a nonspinning black hole), and the streaming patterns differ even more.

For Gargantua the streaming (Figure 8.5) reveals two Einstein rings, shown as pink curves. Outside the outer ring, the stars stream rightward (for example, along the two red curves), as they did for a nonspinning black hole in Figure 8.4. However, the whirl of space has concentrated the stream into narrowed high-speed strips along the back edge of the hole's shadow, strips that bend somewhat sharply at the equator. The whirl has also produced eddies in the streaming (the closed red curves).